Mepivacaine: Drug information - UpToDate

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Contributor Disclosures

For additional information see "Mepivacaine: Patient drug information" and "Mepivacaine: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Polocaine;
- Polocaine Dental;
- Polocaine-MPF;
- Scandonest 3% Plain

Brand Names: Canada

- Carbocaine;
- Polocaine

Pharmacologic Category

- Local Anesthetic

Dosing: Adult

Dosage guidance:

Dosing: Use the smallest dose and concentration required to produce the desired result. Dosage varies with anesthetic procedure, area to be anesthetized, vascularity of the tissues, number of neuronal segments to be blocked, depth of anesthesia and degree of muscle relaxation required, duration of anesthesia, and tolerance and physical condition of the patient. During epidural administration, a test dose (45 to 50 mg) is recommended prior to induction of complete block and all reinforcing doses with continuous catheter technique. The test dose should contain epinephrine to serve as a warning of unintended intravascular injection.

Local or regional anesthesia

Local or regional anesthesia (eg, epidural, caudal, or peripheral nerve blocks): Note: *Maximum daily dose*: 1,000 mg per 24 hours.

Caudal and epidural block (preservative free solutions only): 15 to 30 mL of a 1% solution (maximum total dose: 300 mg) **or** 10 to 25 mL of a 1.5% solution (maximum total dose: 375 mg) **or** 10 to 20 mL of a 2% solution (maximum total dose: 400 mg).

Cervical, brachial, intercostal, pudendal nerve block: 5 to 40 mL of a 1% solution (maximum total dose: 400 mg) **or** 5 to 20 mL of a 2% solution (maximum total dose: 400 mg). For pudendal block, inject one-half the total dose to each side.

Infiltration: Up to 40 mL of a 1% solution (maximum total dose: 400 mg); up to 50 mL of a 1% solution if epinephrine has been added (maximum total dose: 500 mg) (Gropper 2020); an equivalent amount of a 0.5% solution (prepared by diluting the 1% solution with NS) may be used for large areas.

Paracervical block: Up to 20 mL (total for both sides) of a 1% solution (maximum total dose: 200 mg). Inject one-half the total dose to each side. This is the maximum recommended dose per 90-minute procedure; inject slowly with 5 minutes between sides.

Peripheral nerve block to provide a surgical level of anesthesia (off-label):

Major nerve block (blockade of 2 or more distinct nerves, a nerve plexus, or very large nerves at more proximal sites): 30 to 50 mL of a 1% or 1.5% solution (maximum total dose: 350 mg; if epinephrine has been added, maximum total dose: 500 mg) (Ref).

Minor nerve block (blockade of a single nerve [eg, ulnar or radial]): 5 to 20 mL of a 1% solution (maximum total dose: 200 mg) (Ref).

Therapeutic block: 1 to 5 mL of 1% solution (maximum total dose: 50 mg) **or** 1 to 5 mL of 2% solution (maximum total dose: 100 mg).

Transvaginal block (paracervical plus pudendal): Up to 30 mL (total for both sides) of a 1% solution (maximum total dose: 300 mg). Inject one-half the total dose to each side.

Dental anesthesia

Dental anesthesia:

Single site in upper or lower jaw: 51 mg as a 3% solution.

Infiltration and nerve block of entire oral cavity: 270 mg as a 3% solution. Maximum total dose: 400 mg.

Table 1: **Amount of Mepivacaine (3%) Dental Anesthetic per Number of Dental Cartridges (1.7 mL)**

Number of cartridges (1.7 mL)	Mepivacaine mg (3%)
1	51
2	102
3	153
4	204
5	255
6	306
7	357
8	408

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. However, need for dosage adjustment unlikely due to limited duration of use.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling; consider a reduced dose (especially with repeated dosing) in patients with moderate to severe impairment.

Dosing: Older Adult

Refer to adult dosing; reduce dose consistent with age and physical status.

Dosing: Pediatric

(For additional information see ”Mepivacaine: Pediatric drug information”)

Note: Dose varies with procedure, degree of anesthesia needed, vascularity of tissue, duration of anesthesia required, and physical condition of patient. The smallest dose and concentration required to produce the desired effect should be used. Consider incremental administration with negative aspiration prior to each injection; however, absence of blood in the syringe does not guarantee that intravascular injection has been avoided (Ref). Should only be administered under the supervision of a qualified physician experienced in the use of anesthetics.

Dental anesthesia

Dental anesthesia: Children and Adolescents: 3% solution: Injection:

Manufacturer’s labeling: Maximum dose: 5 to 6 mg/kg; maximum total dose: 270 mg. In adults, the dose for single site in upper or lower jaw is 51 mg (one 1.7 mL cartridge).

Alternate dosing: American Academy Pediatric Dentistry (Ref): Maximum dose: 4.4 mg/kg; maximum total dose: 300 mg in any single dental sitting.

Local or regional anesthesia

Local or regional anesthesia (eg, epidural, caudal, or peripheral nerve blocks): Maximum single or total dose given for one procedure: 5 to 6 mg/kg (maximum adult dose per manufacturer: 400 mg); only concentrations <2% should be used in patients <3 years or <14 kg to ensure adequate drug volume for area and to decrease the potential for local anesthetic systemic toxicity.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer’s labeling; use with caution.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer’s labeling; use with caution. Patients with severe hepatic disease, because of their inability to metabolize local anesthetics normally, are at a greater risk of developing toxic plasma concentrations.

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified. Reactions listed may be based on reports for other agents in this same pharmacologic class and may not be specifically reported for mepivacaine.

Postmarketing:

Cardiovascular: Bradycardia, cardiac insufficiency, cardiovascular depression, cardiovascular stimulation, heart block, hypertension, hypotension, low cardiac output, ventricular arrhythmia

Gastrointestinal: Fecal incontinence, loss of anal sphincter control, nausea, vomiting

Genitourinary: Sexual disorder (sexual function), urinary incontinence, urinary retention

Hematologic & oncologic: Methemoglobinemia

Hypersensitivity: Hypersensitivity reaction (including anaphylactic shock, anaphylaxis, angioedema, and nonimmune anaphylaxis) (Estrada 2011; Nam 2023; Takahashi 2019)

Infection: Septic meningitis

Nervous system: Anxiety, asthenia, central nervous system depression, chills, confusion, cranial nerve palsy, dizziness, drowsiness, excitement, headache, loss of consciousness, meningism, paralysis, paresthesia, persistent anesthesia, restlessness, seizure, sensation disorder (loss of perineal sensation), tremor

Neuromuscular & skeletal: Back pain

Ophthalmic: Blurred vision, miosis

Otic: Tinnitus

Respiratory: Apnea, hypoventilation, respiratory depression, respiratory paralysis

Contraindications

Hypersensitivity to mepivacaine, other amide-type local anesthetics, or any component of the formulation

Warnings/Precautions

Concerns related to adverse effects:

• CNS toxicity: Careful and constant monitoring of the patient's state of consciousness should be done following each local anesthetic injection; at such times, restlessness, anxiety, tinnitus, dizziness, blurred vision, tremors, depression, or drowsiness may be early warning signs of CNS toxicity. Use extreme caution in patients with existing neurological disease. Seizures due to systemic toxicity leading to cardiac arrest have also been reported, presumably following unintentional intravascular injection. In the event of cardiovascular collapse and/or severe CNS toxicity, treatment in accordance with the American Society of Regional Anesthesia and Pain Medicine's Checklist for Treatment of Local Anesthetic Toxicity is recommended (ASRA [Neal 2021]).

• Familial malignant hyperthermia: Although product labeling suggests that mepivacaine may potentially trigger malignant hyperthermia, evidence and expert consensus do not support and use is considered to be safe (MHAUS 2023).

• Intra-articular infusion related chondrolysis: Continuous intra-articular infusion of local anesthetics after arthroscopic or other surgical procedures is **not** an approved use; chondrolysis (primarily in the shoulder joint) has occurred following infusion, with some cases requiring arthroplasty or shoulder replacement.

• Methemoglobinemia: Has been reported with local anesthetics; clinically significant methemoglobinemia requires immediate treatment along with discontinuation of the anesthetic and other oxidizing agents. Onset may be immediate or delayed (hours) after anesthetic exposure. Patients with glucose-6-phosphate dehydrogenase deficiency, congenital or idiopathic methemoglobinemia, cardiac or pulmonary compromise, exposure to oxidizing agents or their metabolites, or infants <6 months of age are more susceptible and should be closely monitored for signs and symptoms of methemoglobinemia (eg, cyanosis, headache, rapid pulse, shortness of breath, lightheadedness, fatigue).

• Respiratory arrest: Local anesthetics have been associated with rare occurrences of sudden respiratory arrest. Careful and constant monitoring of the patient's respiratory (adequacy of ventilation) vital signs should be done following each local anesthetic injection.

• Seizures: Convulsions due to systemic toxicity leading to cardiac arrest have also been reported, presumably following unintentional intravascular injection.

Disease-related concerns:

• Cardiovascular disease: Use with caution in patients with cardiovascular disease, including rhythm disturbances, hypotension/shock, and heart block; consider a reduced dose.

• Hepatic impairment: Use with caution in patients with hepatic impairment. Because of the inability to metabolize local anesthetics, patients with severe hepatic disease are at a greater risk of developing toxic plasma concentrations.

• Kidney impairment: Use with caution in patients with kidney impairment.

Special populations:

• Acutely ill patients: Use with caution in acutely ill patients; reduce dose consistent with age and physical status.

• Debilitated patients: Use with caution in debilitated patients; reduce dose consistent with age and physical status.

• Older adult: Use with caution in older adults; reduce dose consistent with age and physical status.

• Pediatric: Use with caution in children; reduce dose consistent with age and physical status.

Dosage form specific issues:

• Preservative-containing solutions: Do not use solutions containing preservatives for caudal or epidural block.

Other warnings/precautions:

• Administration: Intravascular injections should be avoided; aspiration should be performed prior to administration; the needle must be repositioned until no return of blood can be elicited by aspiration; however, absence of blood in the syringe does not guarantee that intravascular injection has been avoided. Use with caution when there is inflammation and/or sepsis in the region of the proposed injection.

• Appropriate dosing: To avoid serious adverse effects and high plasma levels, the lowest dosage resulting in effective anesthesia should be administered. Repeated doses may cause significant increases in blood levels with each repeated dose due to the possibility of accumulation of the drug or its metabolites. Tolerance to elevated blood levels varies with patient status.

• Test dose: A test dose is recommended prior to epidural administration and all reinforcing doses with continuous catheter technique.

• Trained personnel: Clinicians using local anesthetic agents should be well trained in diagnosis and management of emergencies that may arise from the use of these agents. Resuscitative equipment, oxygen, and other resuscitative drugs should be available for immediate use.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as hydrochloride:

Polocaine: 1% (50 mL); 2% (50 mL) [contains methylparaben]

Solution, Injection, as hydrochloride [preservative free]:

Polocaine-MPF: 1% (30 mL); 1.5% (30 mL); 2% (20 mL) [methylparaben free]

Solution, Injection, as hydrochloride [dental use]:

Polocaine Dental: 3% (1.7 mL)

Scandonest 3% Plain: 3% (1.7 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Polocaine Injection)

1% (per mL): $0.38

2% (per mL): $0.39

Solution (Polocaine-MPF Injection)

1% (per mL): $0.45

1.5% (per mL): $0.69

2% (per mL): $0.85

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Administration: Adult

Not for intrathecal use. Intravascular or subarachnoid injections should be avoided. Have resuscitative equipment and medications readily available during administration. Administer slowly in small, incremental doses; avoid rapid injection of a large volume. Use an indwelling IV catheter during major regional nerve block (eg, brachial plexus, lower extremity). Aspirate for blood or cerebrospinal fluid (if applicable) before administration of all doses; absence of blood or cerebrospinal fluid in the syringe does not guarantee that intravascular or intrathecal injection has been avoided. Avoid puncturing the skin when there is inflammation and/or infection in the region of the proposed injection. Do not use injections containing preservatives (eg, methylparaben in multiple-dose vials) for epidural or caudal anesthesia.

Administration: Pediatric

Parenteral: Before injecting, withdraw syringe plunger to ensure injection is not into vein or artery. Administer slowly in small incremental doses, with frequent aspirations before and during the injection to avoid intravascular injection. Use with caution when there is inflammation and/or sepsis in the region of the proposed injection.

Dental: Aspirate the syringe after tissue penetration and before injection to minimize chance of direct vascular injection.

Storage/Stability

Mepivacaine 1%, 1.5%, and 2%: Store at 20°C to 25°C (68°F to 77°F); excursions permitted between 15°C to 30°C (59°F to 68°F). Solutions may be sterilized.

Mepivacaine 3% (dental products): Store at <25°C (77°F); do not freeze. Protect from light.

Use: Labeled Indications

Dental anesthesia: Production of local anesthesia for dental procedures by infiltration or nerve block in adult and pediatric patients.

Local or regional anesthesia (eg, epidural, caudal, or peripheral nerve blocks): Production of local or regional analgesia and anesthesia by local infiltration, peripheral nerve block techniques, and central neural techniques (epidural and caudal) in adult and pediatric patients. **Not** for use in spinal anesthesia.

Medication Safety Issues

Sound-alike/look-alike issues:

Mepivacaine may be confused with BUPivacaine

Polocaine may be confused with prilocaine

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (epidural and intrathecal medications) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Articaine: May increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

BUPivacaine (Liposomal): Local Anesthetics may increase adverse/toxic effects of BUPivacaine (Liposomal). Management: Liposomal bupivacaine should not be administered with local anesthetics, but may be administered 20 minutes or more after lidocaine. Avoid all local anesthetics within 96 hours after administration of liposomal bupivacaine. *Risk X: Avoid*

BUPivacaine: Local Anesthetics may increase adverse/toxic effects of BUPivacaine. Management: Avoid using any additional local anesthetics within 96 hours after insertion of the bupivacaine implant (Xaracoll) or bupivacaine and meloxicam periarticular solution (Zynrelef) or within 168 hours after subacromial infiltration (Posimir brand). *Risk C: Monitor*

Hyaluronidase: May increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

Methemoglobinemia Associated Agents: May increase adverse/toxic effects of Local Anesthetics. Specifically, the risk for methemoglobinemia may be increased. *Risk C: Monitor*

Neuromuscular-Blocking Agents: Local Anesthetics may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Propranolol: May increase serum concentrations of Mepivacaine. *Risk C: Monitor*

PyRIDostigmine: May increase or decrease neuromuscular-blocking effects of Local Anesthetics. *Risk C: Monitor*

ROPivacaine: May increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

Technetium Tc 99m Tilmanocept: Coadministration of Local Anesthetics and Technetium Tc 99m Tilmanocept may alter diagnostic results. Management: Avoid mixing and simultaneously co-injecting technetium Tc 99m tilmanocept with local anesthetics. This interaction does not appear to apply to other uses of these agents in combination. *Risk C: Monitor*

Pregnancy Considerations

Animal reproduction studies have not been conducted. Mepivacaine has been used in obstetrical analgesia.

Breastfeeding Considerations

It is not known if mepivacaine is present in breast milk. The manufacturer recommends that caution be exercised when administering mepivacaine to breastfeeding women. Usual infiltration doses of mepivacaine dental anesthetic given to breastfeeding mothers has not been shown to affect the health of the breastfeeding infant.

Monitoring Parameters

Cardiovascular and respiratory status; vital signs (eg, BP, heart rate, oxygen saturation); mental status changes; signs of CNS toxicity (eg, seizures, visual changes, muscle twitching); systemic toxicity, especially in patients with moderate to severe hepatic or kidney impairment.

Mechanism of Action

Mepivacaine is an amide local anesthetic similar to lidocaine. Local anesthetics bind selectively to the intracellular surface of sodium channels to block influx of sodium into the axon. As a result, depolarization necessary for action potential propagation and subsequent nerve function is prevented. The block at the sodium channel is reversible. When drug diffuses away from the axon, sodium channel function is restored and nerve propagation returns.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action (route and dose dependent): Range: 3 to 20 minutes; Dental: Upper jaw: 30 to 120 seconds; Lower jaw: 1 to 4 minutes

Duration (route and dose dependent): 2 to 2.5 hours; Dental: Upper jaw: 20 minutes; Lower jaw: 40 minutes

Protein binding: ~75%

Metabolism: Primarily hepatic via N-demethylation, hydroxylation, and glucuronidation

Half-life elimination: Neonates: 8.7 to 9 hours; Adults: 1.9 to 3.2 hours

Excretion: Urine (90% to 95% as metabolites)

Patient Counseling Points

(For additional information see ”Mepivacaine: Patient drug information”)

What is this drug used for?

• It is used to numb an area before a procedure.

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

For all uses of this drug:

• Too much acid in the blood (acidosis) like confusion; fast breathing; fast heartbeat; a heartbeat that does not feel normal; very bad stomach pain, upset stomach, or throwing up; feeling very sleepy; shortness of breath; or feeling very tired or weak

• Methemoglobinemia like a blue or gray color of the lips, nails, or skin; a heartbeat that does not feel normal; seizures; severe dizziness or passing out; severe headache; feeling very sleepy; feeling tired or weak; or shortness of breath

• Fast, slow, or abnormal heartbeat

• Restlessness

• Anxiety

• Change in speech

• Feeling lightheaded, sleepy, confused, or having blurred eyesight

• Numbness or tingling in the mouth

• Metallic taste

• Dizziness or passing out

• Ringing in ears

• Shakiness

- Twitching
- Seizures
- Depression
- Trouble breathing, slow breathing, or shallow breathing
- Feeling nervous and excitable
- Severe upset stomach or throwing up
- Feeling hot or cold
- Sneezing
- Sweating a lot
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Epidural:

- Not able to get or keep an erection
- Long-lasting burning, numbness, tingling, or paralysis in the lower half of the body
- Not able to control stools or urine
- Trouble passing urine
- Headache
- Backache
- Fever or chills
- Stiff neck
- If bright lights bother your eyes

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (QA) Qatar: Isocaine | Scandinibsa | Scandonest

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